

Tesla Stock Price Predictions by Using Deep Learning (SimpleRNN and LSTM)

Project Overview

This project focuses on forecasting Tesla stock prices using Deep Learning techniques, specifically Simple Recurrent Neural Networks (SimpleRNN) and Long Short-Term Memory (LSTM) networks. The objective is to analyze historical Tesla stock market data and predict future stock prices based on past market behavior.

The project covers the complete Machine Learning lifecycle, including data cleaning, preprocessing, feature engineering, exploratory data analysis, time-series sequence generation, model development, training, evaluation, and prediction.

Project Workflow ☒ Data Cleaning

The dataset was cleaned to improve data quality by:

☒ Checking and removing duplicate records ☒ Handling missing values ☒ Converting the Date column to datetime format ☒ Setting Date as the index for time-series analysis 2. Data Preprocessing

Data pre-processing involved:

☒ Selecting relevant features ☒ Sorting data chronologically ☒ Preparing the dataset for time-series forecasting 3. Feature Engineering

Additional features were created to capture stock behavior:

☒ Daily Price Range (High – Low) ☒ Daily Returns ☒ Moving Averages ☒ Rolling Statistics ☒ Volume-based indicators

These engineered features help capture market trends and improve prediction accuracy.

4. Data Visualization

☒ Exploratory Data Analysis (EDA) was performed using Matplotlib and Seaborn.

Visualizations included:

☒ Tesla stock price trend analysis ☒ Trading volume analysis ☒ Correlation heatmap ☒ Moving average analysis ☒ Daily return distribution ☒ Actual vs Predicted stock price visualization

These visualizations helped identify trends, volatility patterns, and relationships among features.

◇ Data Scaling

To improve neural network convergence, MinMaxScaler was applied:

Stock price values were normalized between 0 and 1
Reduced the impact of varying feature scales
Improved training stability and convergence speed ◇ Time-Series Sequence Generation

The dataset was transformed into sequential data suitable for recurrent neural networks.

⚡ Approach:

Sliding Window Technique
Past 60 trading days used as input
Next day's Adjusted Closing Price used as target output

🧠 Model Development SimpleRNN Architecture

◇ The SimpleRNN model was developed using TensorFlow Keras Sequential API.

Layers:

◇ SimpleRNN Layer ◇ Dropout Layer ◇ Dense Output Layer

Purpose:

Capture short-term sequential dependencies
Predict future stock prices based on historical data
LSTM Architecture

◇ The LSTM model was developed to overcome limitations of traditional RNNs.

Layers:

◇ LSTM Layer ◇ Dropout Layer ◇ Dense Output Layer

Benefits:

Captures long-term dependencies
Handles vanishing gradient issues
Improves forecasting performance for time-series data
Model Compilation

The models were compiled using:

Loss Function

◇ Mean Squared Error (MSE)

◇ Optimizers ◇ Adam Optimizer ◇ SGD Optimizer ◇ Mini-Batch Gradient Descent

These optimizers were used to minimize prediction errors during training.

🧠 Model Training

⚡ Training strategies included:

◇ Early Stopping ◇ Prevents overfitting ◇ Stops training when validation loss no longer improves 🧠 Model Checkpoint ◇ Saves the best-performing model ◇ Ensures optimal model selection ◇ Training Parameters ◇ Batch Size ◇ Epochs ◇ Validation Split

were tuned for optimal performance.

⚡ Model Evaluation

Model performance was evaluated using:

Mean Squared Error (MSE)

Measures average prediction error.

Visualization

⚡ Actual vs Predicted stock prices were plotted using Matplotlib to visually assess forecasting performance.

Evaluation Steps:

Generate predictions on test data Inverse transform scaled values Compare predictions against actual stock prices Calculate evaluation metrics.

🌐 Streamlit Deployment

Deployment Features

✓ Upload Custom Tesla Stock Dataset ✓ Historical Price Visualization ✓ Trading Volume Analysis ✓ Automated Time-Series Sequence Generation ✓ LSTM-Based Stock Price Prediction ✓ Actual vs Predicted Price Comparison ✓ Real-Time Model Evaluation using MSE ✓ Interactive and User-Friendly Interface

The Streamlit application transforms the Tesla Stock Prediction model into an accessible decision-support tool that can be used for:

Investment Analysis Market Trend Forecasting Financial Research Educational Demonstrations Portfolio Monitoring Deep Learning Model Evaluation

By integrating Deep Learning with an interactive web interface, users can explore historical market trends and evaluate predictive insights without requiring knowledge of the underlying model implementation.

📊 Business Use Cases:

1. Stock Market Trading & Investment Strategies ◇ Automated Trading: ● Use the model's predictions to develop an algorithmic trading strategy. ● Automate buying/selling stocks based on predicted price trends. ◇ Risk Management & Portfolio Optimization: ● Investors can assess potential future price movements to adjust their portfolio allocations. ● Predicting stock volatility helps in hedging risks with options and futures trading.

2. Financial Forecasting & Time-Series Analysis ◇ Long-Term Investment Planning: ● Predict future stock trends for retirement funds, ETFs, or mutual funds. ● Helps in making data-driven decisions on holding or selling assets. ◇ Macroeconomic Analysis: ● Compare Tesla's stock trends with economic indicators like interest rates, inflation, and industry trends.
3. Business & Corporate Use Cases ◇ Company Valuation & Earnings Prediction: ● Tesla can use similar models internally to forecast revenue and profit trends. ● Helps in financial reporting and investor guidance. ◇ Competitor Analysis: ● Apply the model to other stocks (e.g., Rivian, NIO, Lucid Motors) to compare Tesla's growth with competitors.



Conclusion

This project demonstrates the application of Deep Learning techniques for Tesla stock price forecasting. Both SimpleRNN and LSTM models were developed and evaluated using historical stock market data. The results indicate that recurrent neural networks can effectively capture temporal patterns and trends, making them valuable tools for financial forecasting. Future enhancements involving advanced architectures such as GRU and Transformers can further improve prediction accuracy and robustness.